Mechanics Cheat Sheet

1. Work

- **Definition:** Work done by a force $\vec{F}$ causing displacement $\vec{d}$.
- **Constant Force:**
$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta$$
- θ: angle between $\vec{F}$ and $\vec{d}$.
- **Special Cases:**
  - $\theta = 0^\circ \Rightarrow W = Fd$ (maximum positive work)
  - $\theta = 90^\circ \Rightarrow W = 0$
  - $\theta = 180^\circ \Rightarrow W = -Fd$ (maximum negative work)
- **Variable Force:**
$$W = \int_i^f \vec{F} \cdot d\vec{r}$$
- For 1D motion:
$$W = \int_{x_i}^{x_f} F(x)\,dx$$
- **Trick:** Area under the F-x graph gives work done.
- **Units:** Joule (J), erg, electron-volt (eV).
- **Nature:** Scalar quantity.
- **Work-Energy Theorem:**
$$W_{\text{net}} = \Delta K = K_f - K_i$$

2. Energy

- **Definition:** Capacity to do work.
- **Units:** Same as work: J, erg, eV.
- **Nature:** Scalar quantity.

2.1 Kinetic Energy (K)

- Translational kinetic energy:
$$K = \frac{1}{2}mv^2$$
- Relation with momentum:
$$K = \frac{p^2}{2m}, \qquad p = mv$$
- **Trick:** If p is constant, $K \propto \frac{1}{m}$.

2.2 Potential Energy (U)

- **Definition:** Energy due to position or configuration.
- Defined only for conservative forces.
- Relation:
$$\vec{F}_{\text{cons}} = -\nabla U$$
- For 1D:
$$F_x = -\frac{dU}{dx}$$
- Change in potential energy:
$$U_f - U_i = -\int_i^f \vec{F}_{\text{cons}} \cdot d\vec{r}$$

2.3 Gravitational Potential Energy

- Near Earth's surface:
$$U_g = mgh$$
- Universal gravitation:
$$U_g = -\frac{GMm}{r}$$

2.4 Elastic Potential Energy

- For a spring:
$$U_s = \frac{1}{2}kx^2$$
- Hooke's law:
$$F_s = -kx$$

3. Power

- **Definition:** Rate of doing work or transferring energy.
- Average power:
$$P_{\text{avg}} = \frac{\Delta W}{\Delta t}$$
- Instantaneous power:
$$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$$
- **Units:** Watt (W) = J/s

4. Work-Energy and Conservation

- General work-energy relation:
$$W_{\text{total}} = \Delta K$$
- If conservative and non-conservative forces act:
$$W_{\text{cons}} + W_{\text{non-cons}} = \Delta K$$
- Since
$$W_{\text{cons}} = -\Delta U$$
then
$$-\Delta U + W_{\text{non-cons}} = \Delta K$$
- Rearranging:
$$W_{\text{non-cons}} = \Delta K + \Delta U = \Delta E_{\text{mech}}$$
- **Trick:** With friction, $W_{\text{friction}} < 0$ and mechanical energy decreases.
- **Total Energy Conservation:** In an isolated system, total energy remains constant.

5. Collisions

- **Momentum Conservation:**
$$m_1\vec{v}_{1i} + m_2\vec{v}_{2i} = m_1\vec{v}_{1f} + m_2\vec{v}_{2f}$$
- **Coefficient of Restitution:**
$$e = \frac{\text{relative speed of separation}}{\text{relative speed of approach}}$$
- For 1D:
$$e = \frac{v_{2f} - v_{1f}}{v_{1i} - v_{2i}}$$

5.1 Types of Collisions

- **Elastic Collision:**
  - Momentum conserved
  - Kinetic energy conserved
  - $e = 1$
- **Inelastic Collision:**
  - Momentum conserved
  - Kinetic energy not conserved
  - $0 < e < 1$
- **Perfectly Inelastic Collision:**
  - Bodies stick together
  - $e = 0$
  - Common final velocity:
$$(m_1 + m_2)V_f = m_1v_{1i} + m_2v_{2i}$$

6. Vertical Circular Motion

- Minimum speed at top for looping:
$$v_{\text{top}} \geq \sqrt{gR}$$
- Minimum speed at bottom:
$$v_{\text{bottom}} \geq \sqrt{5gR}$$
- At any point:
$$T \text{ or } N = \frac{mv^2}{R} + mg\cos\theta$$
- At top:
$$T \text{ or } N = \frac{mv^2}{R} - mg$$
- At bottom:
$$T \text{ or } N = \frac{mv^2}{R} + mg$$

7. Spring-Mass System

- Hooke's law:
$$F_s = -kx$$
- Elastic potential energy:
$$U_s = \frac{1}{2}kx^2$$
- Simple harmonic motion equation:
$$x(t) = A\cos(\omega t + \phi)$$
- Angular frequency:
$$\omega = \sqrt{\frac{k}{m}}$$
- Period:
$$T = 2\pi\sqrt{\frac{m}{k}}$$

8. Equilibrium and Stability

- **Stable Equilibrium:** Small displacement produces restoring tendency.
- **Condition:** U minimum, $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} > 0$.
- **Unstable Equilibrium:** Small displacement drives the system farther away.
- **Condition:** U maximum, $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} < 0$.
- **Neutral Equilibrium:** Small displacement leaves the system in its new position.
- **Condition:** U constant, $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} = 0$.